Three Tasks, One Animal, One Pipeline

Short-clip behavior classification, detection and feature analysis for a zoo civet



CAMERA 02 · 19 MAY 2026

Project team, CMU

What the brief asked, what we delivered

TASK	BRIEF ASKED	WHAT WE BUILT	MEASURED RESULT
TASK 1 Image-based classification	Walking · Eating · Resting · Climbing	Five-second window CNN, three classes	Macro F1 **0.949** on 15 held-out windows
TASK 2 YOLO-based detection	Video → frames → YOLO annotation → training → detection	Auto-labeled detector, no hand-drawn boxes	mAP50 **0.537** on validation; animal found in 59% of held-out frames
TASK 3 Keypoint feature analysis	Body Angle · Head Angle · Movement Speed · Limb Distance	Silhouette geometry with the same four names	Macro F1 **0.422** on 15 test windows

Climbing dropped with instructor confirmation (ADR 0007). Task 3 is a silhouette proxy, not keypoints (ADR 0009).

Task 1: the evidence base

22	201	138	93 · 30 · 15
source clips	five-second windows	labelled windows	train · val · test

01
Source clips
22 fixed-camera MP4s

→

02
Verified intervals
one class per whole span

→

03
Five-second windows
a fixed grid over each clip

→

04
Split by whole clip
no frame-level leakage

THE TEST SPLIT

The eight-clip gold set: two labeling passes, interval Dice **0.597**, reconciled, and locked on 2026-10-20 before the test was read.

63 of 201 windows are dropped: 46 not covered completely by one interval, 17 touching a Transition or Uncertain span.

Task 1: 0.949 held-out macro F1

0.949

MACRO F1 · 15 HELD-OUT WINDOWS

ONE ERROR

Clip 25690519193808204, 40–45 s.
Labelled Eating; predicted Resting at 0.847.

CONFUSION MATRIX · ROWS TRUE / COLUMNS PREDICTED

	EATING	LOCOMOTION	RESTING	TOTAL
Eating	6	0	1	7
Locomotion	0	2	0	2
Resting	0	0	6	6
TOTAL	6	2	7	15

Abstained: 0 of 15. Accuracy over decided windows: 0.933.

WHAT THIS NUMBER DOES NOT ESTABLISH

Fifteen windows: Locomotion has 2, Resting takes all 6 from one clip, and camera and class travel together. 0.949 is an upper bound on a new recording, not an estimate of it.

The one error may be a model miss on a partly hidden animal or an annotation error; the label was not changed after the test was read.

Task 2: nobody drew a box

METHOD TRIED	WHAT IT RETURNED
COCO YOLO11n, full frame	Zero detections of the animal. One potted plant at 0.344.
Frame differencing, largest blob	Locks onto swaying foliage, not the animal.
Grounding DINO, open vocabulary	40+ boxes at 0.12; top score 0.535 on empty floor.
Per-pixel median background	408 boxes kept after a one-second persistence filter.



YOLO BOX 0.66 · CAMERA 03 · 19 MAY 2026
Source: outputs/detection/heldout.csv

The cameras never move, so each clip supplies its own background.

Median frame per clip · per-pixel noise threshold · components in an animal-size band · a box must persist for about a second.

Task 2: the loop that learns its own labels

01
1061
frames sampled at 1 fps across 22 clips

02
470
candidate boxes from the background model

03
408
kept after the one-second persistence filter

04
116
boxes audited; 113 were on the animal

05
0.537
YOLO round 1 mAP50; round 2 self-trains

0.499
mAP50 on untouched validation after self-training

59%
of held-out frames fire at confidence 0.25

5 / 10
audited held-out boxes were on the animal

ROUND 2 ON ITS OWN VALIDATION: 0.774

That number is not comparable: 115 of round 2's validation labels were partly written by round 1.

No person drew a bounding box at any point.

Task 3: four features from a silhouette

FEATURE · MEDIAN PER CLASS	EATING	LOCOMOTION	RESTING
Body angle (deg)	115°	88°	154°
Head angle (deg)	113°	49°	96°
Movement speed (px/s)	17	134	2
Limb extent (px)	56	148	98

Medians over training and validation windows with at least one silhouette.

VALIDATION MACRO F1

0.479

TEST MACRO F1

0.422

PROXY, NOT POSE ESTIMATION

No keypoints, no pose model. Resting is invisible: 58 of 93 training windows have no silhouette, so Resting scores 0 on test.

Per-class test F1: Eating 0.267 · Locomotion 1.000 · Resting 0.000. Locomotion separates by an order of magnitude in speed.

All three tasks in one command

```
$ python3 scripts/demo_all.py data/raw/Eating/25690519193422801.mp4
```

```
data/raw/Eating/25690519193422801.mp4  (50.0 s, 10 windows, abstain below 0.00)
  start     end  T1           conf  T2 box               via     conf  crop         body  head   speed  limb
   0.00    5.00  Eating       0.73  408,462,606,528      motion  0.00  Eating      6 161       1    63
   5.00   10.00  Eating       0.69  606,510,674,564      motion  0.37  Eating      25 16      23    52
  10.00   15.00  Eating       0.69  632,472,700,550      motion  0.00  Eating      43 24      20    68
  15.00   20.00  Eating       0.67  341,365,624,534      yolo    0.58  Locomotion  119 19      24    51

detection: 10/10 windows found the animal directly (6 by the trained detector, 4 by the per-clip model)
full-frame and crop labels agree on 4/10 windows
```

Task 1 behavior, Task 2 detector box and crop, Task 3 features — same windows, same timecodes.

Validation clip. Output trimmed to the header, four data rows and the two summary lines; the per-class features are medians over the window's sampled frames.

What this does not prove

01	15 test windows from 5 clips	Locomotion has 2 and Resting takes 6 from one clip. A per-class score on that support is a baseline, not a stable estimate.
02	Camera and class travel together	Every Resting window is Camera 03 and nearly all Eating is Camera 02, so a score can come from the recording rather than the behavior.
03	The detector does not generalize	It fails on the 2026-05-20 recording day, absent from training, and fires on a bright plant.
04	Task 3 is a proxy	No keypoints and no pose model. A still animal joins the median background, so most Resting windows have no features at all.
05	One training seed	Seed 0 only. No run-to-run spread is reported, so the split between score and seed is unknown.

And no human ground truth for detection: its quality is a counted visual audit, not mAP against truth.

Next

Four changes that would turn these numbers into evidence.

01 Hold out a recording group

Collect footage from sessions absent from training and repeat the evaluation, so the upper bound becomes an estimate.

02 Widen per-class support

Add Locomotion and Resting test clips from more than one clip per class before reading per-class differences.

03 Calibrate abstention

Choose the abstain threshold on a larger validation set or a separate calibration set, never on the test windows.

04 Revisit Climbing and pose

Only if verified footage and keypoint labels exist. Until then, three classes and silhouette features are the scope.

A reproducible baseline that states what it measured and what it did not.